

Coimisiún na Scrúduithe Stáit

State Examinations Commission

Leaving Certificate 2026

Marking Scheme

Mathematics

Foundation Level

Note to teachers and students on the use of published marking schemes

Marking schemes published by the State Examinations Commission are not intended to be standalone documents. They are an essential resource for examiners who receive training in the correct interpretation and application of the scheme. This training involves, among other things, marking samples of student work and discussing the marks awarded, so as to clarify the correct application of the scheme. The work of examiners is subsequently monitored by Advising Examiners to ensure consistent and accurate application of the marking scheme. This process is overseen by the Chief Examiner, usually assisted by a Chief Advising Examiner. The Chief Examiner is the final authority regarding whether or not the marking scheme has been correctly applied to any piece of candidate work.

Marking schemes are working documents. While a draft marking scheme is prepared in advance of the examination, the scheme is not finalised until examiners have applied it to candidates' work and the feedback from all examiners has been collated and considered in light of the full range of responses of candidates, the overall level of difficulty of the examination and the need to maintain consistency in standards from year to year. This published document contains the finalised scheme, as it was applied to all candidates' work.

In the case of marking schemes that include model solutions or answers, it should be noted that these are not intended to be exhaustive. Variations and alternatives may also be acceptable. Examiners must consider all answers on their merits, and will have consulted with their Advising Examiners when in doubt.

Future Marking Schemes

Assumptions about future marking schemes on the basis of past schemes should be avoided. While the underlying assessment principles remain the same, the details of the marking of a particular type of question may change in the context of the contribution of that question to the overall examination in a given year. The Chief Examiner in any given year has the responsibility to determine how best to ensure the fair and accurate assessment of candidates' work and to ensure consistency in the standard of the assessment from year to year. Accordingly, aspects of the structure, detail and application of the marking scheme for a particular examination are subject to change from one year to the next without notice.

Coimisiún na Scrúduithe Stáit
State Examinations Commission

Leaving Certificate Examination 2026

Mathematics

Foundation Level

Marking Scheme

300 marks

General notes on the marking

In general, accept a candidate's work in one part of a question for use in subsequent parts of the question, unless this oversimplifies the work involved.

Scales and Slips

Certain types of mistake – typically involving incorrect rounding, omission of units, a misreading that does not oversimplify the work, or an arithmetical error that does not oversimplify the work – are referred to as a **slip** and are denoted with a *. In such cases, a mark that is one mark below the full-credit mark may be awarded. This level of credit is referred to as *Full Credit –1*. Thus, for example, in Scale 10C, *Full Credit –1* of 9 marks may be awarded.

The only marks that may be awarded for a question are those on the appropriate scale in the marking scheme, or *Full Credit –1*.

Instructions regarding penalties for omitted or incorrect units are given in the scheme for each question to which they apply. A penalty for rounding is applied once per unit of marking (i.e., if parts (a) and (b) are marked as a single unit, a penalty for rounding is only applied once for (a) and (b) combined).

Steps

Where steps are listed in the Marking Notes, unless otherwise specified, it is to be taken that they can be independently correct / incorrect – that is, in a candidate's solution, step n can be considered correct even if previous step(s) have not been correctly presented, as long as the work done to arrive at step n from the previous step(s) has not been oversimplified. Note also that these steps may not need to be presented in the order specified in the Marking Notes. In general, it is specifically noted where these do not hold.

Errors

An error is a mistake in a candidate's work that is not considered a slip. Where a question is **not** marked using steps, if a candidate has a single error, they are generally awarded one level of credit below that which they would otherwise have been awarded. Similarly, where they have two errors, they are generally awarded two levels of credit below that which they would otherwise have been awarded. (If they present sufficient work for *Low Partial Credit*, they will be awarded this at a minimum, regardless of the number of errors.) For example, on a C-scale:

- *High Partial Credit*: One error, otherwise fully correct (or fully correct with a *)
- *Low Partial Credit*: Two errors, otherwise fully correct (or fully correct with a *)

Where a question **is** marked using steps, this does not apply. Instead, an error in a step means that the step has not been completed correctly; this does not affect the completion of other steps (unless it oversimplifies the work). So, if a candidate has multiple errors on a single step, they could still be awarded up to *High Partial Credit*, depending on the marking scheme.

Slips and Errors

Where a candidate has a single * on their solution, this is ignored in the awarding of credit unless they would otherwise have *Full Credit*. If they would otherwise have *Full Credit*, then *Full Credit –1* is awarded. Where a candidate has multiple *s, this is generally treated as an error. This applies even where the *s are in different steps.

Multiple answers
Where a solution requires substantial work, all separate attempts are marked and the marks for the best one are awarded, regardless of crossing out.

Where a solution requires selection from the question:

- If a candidate has only one answer, which they have then crossed out, the crossing-out is ignored
- If a candidate has both crossed-out answer(s) and answer(s) that are not crossed out, ignore the crossed-out answer(s) and only mark those that are not crossed out
- If a candidate has multiple answers that are **not** crossed out, award the lowest mark associated with these answers (generally, such a set of answers will end up being considered incorrect).

Excess work
Where a solution requires substantial work, incorrect excess work (i.e. work presented after the answer) is generally not penalised. This does not apply where candidates are asked to write a sentence, or number of sentences, to explain / justify / describe something. In such cases, the answer as a whole is considered.

Where a subsequent part of a question requires candidates to use their answer from a previous part, and an incorrect answer arising from incorrect excess work is used, this is generally treated as an error in the subsequent part, unless it oversimplifies the work involved (in which case a lower level of credit is generally awarded for the subsequent part).

Square brackets
Where something is contained in square brackets in the model solution, it is **not** required for *Full Credit*.

Work of merit
Where the scheme indicates "work of merit" (or "substantial work of merit"), examples are given to exemplify the standard of work required to be considered work of merit (or substantial work of merit) in that particular question.

Accept
Where it is indicated to "accept work, this means that, for this year, such work is to be accepted for *Full Credit*, unless otherwise indicated.

Palette of annotations available to examiners

Symbol	Name	Meaning in the body of the work	Meaning when used in the right margin
✓	Tick	Work of relevance	The work presented in the body of the script merits full credit
✗	Cross	Incorrect work (distinct from an error)	The work presented in the body of the script merits 0 credit
*	Star	Rounding / Unit / Arithmetic error Misreading	
	Horizontal wavy	Error	
P	P		The work presented in the body of the script merits a partial credit for B scales
L	L		The work presented in the body of the script merits low partial credit
M	M		The work presented in the body of the script merits mid partial credit
H	H		The work presented in the body of the script merits high partial credit
F*	F star		The work presented in the body of the script merits Full Credit (– 1)
[	Left Bracket		Another version of this solution is presented elsewhere and it merits equal or higher credit
	Vertical wavy	No work on this page (portion of the page)	
WOM		The candidate has presented work or merit	
O	Oversimplify	The candidate has oversimplified the work	

Note: Where work of substance is presented in the body of the script, the annotation on the right margin should reflect a combination of annotations in the work.

In a **C scale** that is **not** marked using steps, where * and  and  appear in the body of the work, then L should be placed in the right margin.

In the case of a **D scale** with the same annotations, M should be placed in the right margin.

Note: Where work is presented for a question in the answerspace provided, and a separate attempt is presented on additional pages, both attempts are marked.

For each attempt, the annotation for the appropriate credit level is put in the right margin next to that attempt. The annotation for the lower credit level is then left-bracketed, and the mark for the higher credit level is awarded.

Structure of the marking scheme

Candidate responses are marked according to different scales, depending on the types of response anticipated. Scales labelled A divide candidate responses into two categories (correct and incorrect). Scales labelled B divide responses into three categories (correct, partially correct, and incorrect), and so on. The scales and the marks that they generate are summarised in this table:

Scale label	**B**	**C**	**D**
No of categories	3	4	5
5-mark scale	0, 2, 5	0, 2, 3, 5	0, 2, 3, 4, 5
10-mark scale	0, 5, 10	0, 4, 7, 10	0, 4, 6, 8, 10
15-mark scale		0, 5, 10, 15	0, 4, 7, 10, 15

A general descriptor of each point on each scale is given below. More specific directions in relation to interpreting the scales in the context of each question are given in the scheme, where necessary.

Marking scales – level descriptors

B-scales (three categories)

- response of no substantial merit (no credit)
- partially correct response (partial credit)
- correct response (full credit)

C-scales (four categories)

- response of no substantial merit (no credit)
- response with some merit (low partial credit)
- almost correct response (high partial credit)
- correct response (full credit)

D-scales (five categories)

- response of no substantial merit (no credit)
- response with some merit (low partial credit)
- response about half-right (mid partial credit)
- almost correct response (high partial credit)
- correct response (full credit)

Model Solutions & Marking Notes

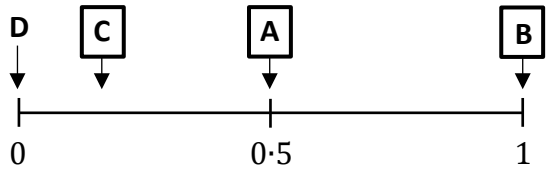
Note: The model solutions for each question are not intended to be exhaustive – there may be other correct solutions. Any Examiner unsure of the validity of the approach adopted by a particular candidate to a particular question should contact his / her Advising Examiner.

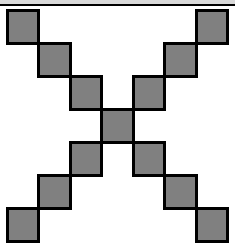
Q1	Model Solution – 30 Marks	Marking Notes
(a) **(i)** **(ii)**	**(i)** $60 + 80 = 140$ **(ii)** $0{\cdot}7 \times 60 = 42$	**Scale 10D (0, 4, 6, 8, 10)** *Low Partial Credit:* • Work of merit, for example, in **(i)**, some relevant addition, mentions 60 and 80; in **(ii)**, some relevant multiplication or mentions $0{\cdot}7$ *Mid Partial Credit:* • One part correct • Work of merit in both parts *High Partial Correct* • One part correct and work of merit in the other part
(a) **(iii)**	$\frac{3}{5} \times 80 = 48$ $80 - 48 = 32$ **OR** $\frac{2}{5} \times 80 = 32$	**Scale 10C (0, 4, 7, 10)** *Low Partial Credit:* • Work of merit, for example, calculates 16; mentions $\frac{2}{5}$ *High Partial Credit:* • Finds 48

Q1	Model Solution – 30 Marks	Marking Notes
(b) **(i)** **(ii)**	**(i)** $\frac{7+3}{5} = \frac{10}{5} = 2$ **(ii)** $(7{\cdot}35 + 2{\cdot}53) \div 4{\cdot}6 =$ **OR** $\frac{7{\cdot}35 + 2{\cdot}53}{4{\cdot}6}$ (or similar)	**Scale 10D (0, 4, 6, 8, 10)** *Note*: Apply a * for each incorrect rounding in part **(i)** *Low Partial Credit:* • Work of merit, for example, in **(i)**, some correct rounding; in **(ii)**, shows some understanding of order of operations *Mid Partial Credit:* • One part correct • Work of merit in both parts *High Partial Correct:* • One part correct and work of merit in the other part

Q2	Model Solution – 30 Marks	Marking Notes
(a)	**(i)** Mode $= 15$ **(ii)** Mean $= \frac{84}{6} = 14$	**Scale 10D (0, 4, 6, 8, 10)** *Low Partial Credit:* • Work of merit, for example, in **(i)**, highlights both 15's, reference to most common; in **(ii)**, some relevant addition; 84 or 6 written • Calculates mean in part **(i)** with supporting work • Calculates median $(14{\cdot}5)$ in part **(i)** and/or part **(ii)** with supporting work *Mid Partial Credit:* • One part correct • Work of merit in both parts • Mode and mean correct and reversed without relevant work *High Partial Correct:* • One part correct and work of merit in the other part • Mode and mean correct and reversed with relevant work

Q2	Model Solution – 30 Marks	Marking Notes
(b)	$1200 \div 200 = 6$ (2 free) $2{\cdot}49 \times 4 = [€]\ 9{\cdot}96$	**Scale 10C (0, 4, 7, 10)** *Low Partial Credit:* • Work of merit, for example, mentions 2 free; calculates 6 containers *High Partial Credit:* • Calculates 4 containers • Correct answer without supporting work • Calculates cost of 3 containers/600 ml as €$4{\cdot}98$
(c)	*Answer*: Small Punnet *Calculations:* $5{\cdot}50 \div 500 = €0{\cdot}011$ (per gram) $2{\cdot}95 \div 300 = €0{\cdot}00983\dot{}$ (per gram) **OR** $500 \div 5{\cdot}50 = 90{\cdot}9\ldots. €1 = 90{\cdot}9$ grams $300 \div 2{\cdot}95 = 101{\cdot}7\ldots. €1 = 101{\cdot}7$ grams **OR** $€2{\cdot}95 \div 300 \times 500 = €4{\cdot}92$ **OR** $€5{\cdot}50 \div 500 \times 300 = €3{\cdot}30$ C	**Scale 10C (0, 4, 7, 10)** *Low Partial Credit:* • Correct answer with incorrect or no calculations • Work of merit in calculations, for example, calculates cost per gram/euro for one brand; calculates the cost for a weight not listed in the table, e.g. 1000g of a large punnet $=$ €11. *High Partial Credit:* • Correct calculations with incorrect answer • Correct answer with work of merit in calculations

Q3	Model Solution – 30 Marks	Marking Notes
(a)	**(i)** Median $= \frac{72+74}{2} = 73$ **(ii)** Range $= 88 - 64 = 24$	**Scale 15D (0, 4, 7, 10, 15)** *Low Partial Credit:* • Work of merit, for example, in **(i)**, some effort to find the middle number in **(ii)**, identifies max or min value *Mid Partial Credit:* • One part correct • Work of merit in both parts *High Partial Correct:* • One part correct and work of merit in the other part
(a) **(iii)**	The hearts rates have increased but the spread is the same.	**Scale 5B (0, 2, 5)** *Partial Credit:* • Work of merit, for example, identifies the heart rate increases or the spread stays the same
(b)	**(i)**  **(ii)** Ben rolls a 7 (or similar)	**Scale 10C (0, 4, 7, 10)** *Low Partial Credit:* • Work of merit, for example, in **(i)**, one correct entry; in **(ii)**, recognises that event is impossible *High Partial Credit:* • One part correct • Work of merit in both parts

<table>
<tr><th>Q4</th><th>Model Solution – 30 Marks</th><th>Marking Notes</th></tr>
<tr><td>(a)</td><td> 
Pattern 4</td><td>Scale 10B (0, 5, 10)

Partial Credit:
• Some work of merit, for example one or more extra box added to pattern 3</td></tr>
<tr><td>(b)
(c)</td><td>(b)
<table><tr><td>Pattern 3</td><td>Pattern 4</td><td>Pattern 5</td><td>Pattern 6</td></tr><tr><td>9</td><td>13</td><td>17</td><td>21</td></tr></table>
(c)
Answer: Linear

Reason: It’s increasing by 4 each term</td><td>Scale 15D (0, 4, 7, 10, 15)

Note: A correct entry in the table is a previous term increased by 4

Low Partial Credit:
• Work of merit, for example,
in (b), a linear pattern created with at least 3 terms; one correct entry;
in (c), answer or reason correct

Mid Partial Credit:
• One part correct
• Work of merit in both parts

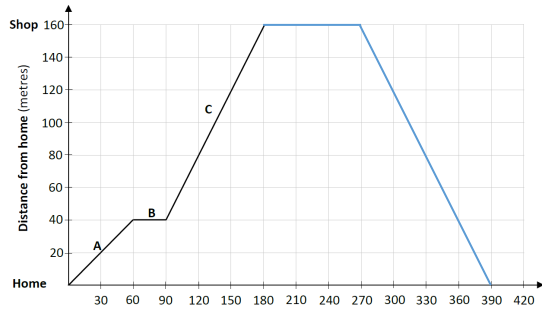
High Partial Credit:
• One part correct and work of merit in the other part</td></tr>
<tr><td>(d)</td><td>(i)
$4(63) - 3$
$= 252 - 3$
$= 249$

(ii)
$4k - 3 = 373$
$4k = 376$
$k = 94$</td><td>Scale 5D (0, 2, 3, 4, 5)
Note: Supporting work must be shown in part (ii) to be considered correct

Low Partial Credit:
• Work of merit in, for example,
in (i), some correct substitution;
in (ii), sets up equation, adds 3 to 373, or divides by 4
• Part (ii) correct without supporting work

Mid Partial Credit:
• One part correct
• Work of merit in both parts

High Partial Credit:
• One part correct and work of merit in the other part</td></tr>
</table>

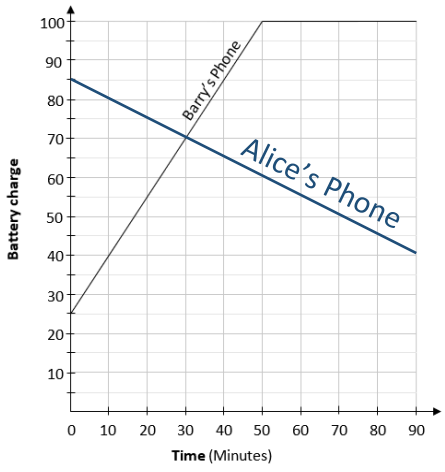
Q5	Model Solution – 30 Marks	Marking Notes
(a) **(b)**	**(a)** **(i)** 40 [m] **(ii)** 30 [seconds] **(b)** *Answer*: C *Reason*: It's steeper (or similar)	**Scale 15D (0, 4, 7, 10, 15)** *Low Partial Credit:* • Work of merit, for example, in **(a)**, gives 60 as answer in **(i)**; mentions 60 or 90 in **(ii)**; in **(b)**, correct answer but no or incorrect reason; some attempt at reason *Mid Partial Credit:* • Part **(a)** or **(b)** correct • Work of merit in both parts **(a)** and **(b)** *High Partial Credit:* • One part correct (**(a)** or **(b)**) and work of merit in the other part
(c) **(i)**		**Scale 5C (0, 2, 3, 5)** *Low Partial Credit:* • Some work of merit *High Partial Credit:* • One correct stage constructed
(c) **(ii)**	$S = \frac{Distance}{Time} = \frac{160}{120} = 1.33$ [m/s] [2 D.P.]	**Scale 10C (0, 4, 7, 10)** *Low Partial Credit:* • Work of merit, for example, writes down the correct formula; identifies the correct distance or time *High Partial Credit:* • Writes down the correct formula, identifies the correct distance and time but fails to substitute into formula • Calculates the speed for the entire journey • Gives answer as $\frac{4}{3}$ • Calculates speed in m/min

<table>
<tr><th>Q6</th><th>Model Solution – 30 Marks</th><th>Marking Notes</th></tr>
<tr><td>(a)</td><td>(i)
25
15

(ii)
His phone was fully charged (or similar)</td><td>Scale 10D (0, 4, 6, 8, 10)

Low Partial Credit:
• Work of merit, for example,
in (i), relevant indication on graph; one number correct;
in (ii), some work of merit

Mid Partial Credit:
• Part (i) or (ii) correct
• Work of merit in both parts

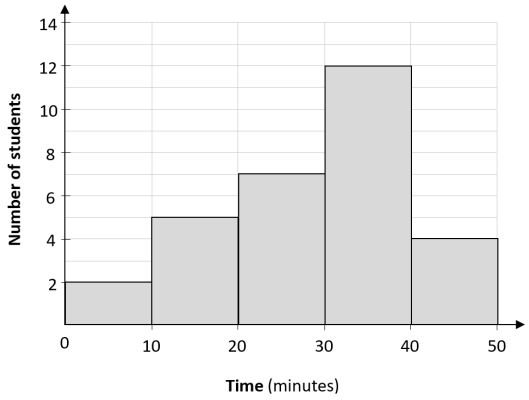
High Partial Credit:
• One part correct and work of merit in the other part</td></tr>
<tr><td>(b)
(i)
(ii)</td><td>(i)
<table><tr><th>Minutes</th><th>Battery charge on Alice’s phone (%)</th></tr><tr><td>0</td><td>85</td></tr><tr><td>10</td><td>80</td></tr><tr><td>20</td><td>75</td></tr><tr><td>30</td><td>70</td></tr><tr><td>40</td><td>65</td></tr><tr><td>50</td><td>60</td></tr><tr><td>60</td><td>55</td></tr></table>(ii)
 
</td><td>Scale 15D (0, 4, 7, 10, 15)

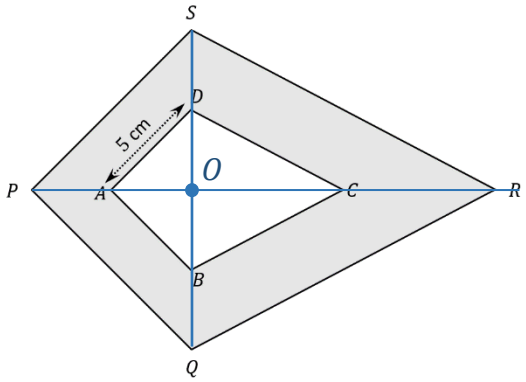
Low Partial Credit:
Work of merit, for example,
in (i), 1 correct entry in table;
in (ii), 1 point correctly graphed

Mid Partial Credit:
• One part correct
• Work of merit in both parts

High Partial Credit:
• One part correct and work of merit in the other part</td></tr>
</table>

Q6	Model Solution – 30 Marks	Marking Notes
(b) **(iii)**	*Answer*: $C = 85 - 0{\cdot}5t$ *Justification*: 85 is the value at $t = 0$ and the value goes down by $0{\cdot}5$ each minute **OR** Shows formula works for 2 values **OR** Other valid reason	**Scale 5B (0, 2, 5)** *Partial Credit:* • Correct answer with no or incorrect reason • Incorrect answer with work of merit in reason given

Q7	Model Solution – 30 Marks	Marking Notes
(a) **(b)**	**(a)** $30 - 23 = 7$ **(b)**  	**Scale 10D (0, 4, 6, 8, 10)** *Low Partial Credit:* • Work of merit, for example, in **(a)**, some relevant addition/subtraction; in **(b)**, one bar correct; correct height indicated • Part **(a)** correct *Mid Partial Credit:* • Part **(a)** correct and work of merit in part **(b)** • Work of merit in both parts *High Partial Credit:* • Part **(b)** correct *Full Credit (-1):* • Heights all correct but one or more bars an incorrect width
(c) **(d)**	**(c)** $2 + 5 = 7$ **(d)** $\frac{4}{30} \times 100 = 13{\cdot}3\ [\%]$ [1 d.p.]	**Scale 10D (0, 4, 6, 8, 10)** *Low Partial Credit:* • Work of merit, for example, in **(c)**, relevant addition, 2 or 5 written; in **(d)**, identifies 4, use of 30; shows some knowledge of percentages *Mid Partial Credit:* • One part correct • Work of merit in both parts *High Partial Correct:* • One part correct and work of merit in the other part
(e)	*Answer*: $30 - 40$ *Reason*: Median is the centre of the 15th and 16th person. There are 14 people in the first three intervals so the next two people lie in the fourth interval.	**Scale 10C (0, 4, 7, 10)** *Low Partial Credit:* • Work of merit in reason • Shows understanding of median being the middle term in justification *High Partial Credit:* • Correct answer with no or incorrect reason

Q8	Model Solution – 30 Marks	Marking Notes
(a) (i) **(ii)**	**(i)** $5 \times 2 = 10$ **cm** **(ii)** $42 \times 2^2 = 168$ [cm^2]	**Scale 10D (0, 4, 6, 8, 10)** *Low Partial Credit:* • Work of merit, for example, in **(i)**, use of scale factor; in **(ii)**, multiplies by scale factor *Mid Partial Credit:* • One part correct • Work of merit in both parts *High Partial Correct:* • One part correct and work of merit in the other part *Full Credit (-1):* • Apply a * for no unit in part **(i)**
(a) (iii)		**Scale 10C (0, 4, 7, 10)** *Low Partial Credit:* • Correct line constructed • O correctly plotted without construction lines • O plotted with incorrect lines *High Partial Credit:* • Two correct lines constructed *Full Credit (-1):* • Apply a * for no label

Q8	Model Solution – 30 Marks	Marking Notes
(b)	**(i)** Opposite $= 50$ Adjacent $= 160$ **(ii)** $\tan A = \frac{50}{160}$ **(iii)** $A = \tan^{-1}\frac{50}{160} = 17{\cdot}354.. = 17[°]$	**Scale 10D (0, 4, 6, 8, 10)** Note: Accept correct answer for part **(ii)** if given in part **(iii)** *Low Partial Credit:* • Work of merit, for example, in **(i)**, 50 or 160 correct; in **(ii)**, one correct value; answer as $\frac{a}{b}$ from correctly labelled diagram in **(iii)**, $\tan^{-1}$ written *Mid Partial Credit:* • One part correct • Work of merit in two parts *High Partial Correct:* • Two parts correct • One part correct and work of merit in other two parts

Q9	Model Solution – 45 Marks	Marking Notes
(a)	**(i)** $3 \times 2 \times 4 = 24$ **(ii)** $2 \times 1 \times 4 = 8$	**Scale 10D (0, 4, 6, 8, 10)** *Low Partial Credit:* • Work of merit in **(i)** or **(ii)**, for example, some relevant multiplication or identified the three relevant numbers *Mid Partial Credit:* • One part correct • Work of merit in both parts *High Partial Correct:* • One part correct and work of merit in the other part
(b)	$370 \times 1{\cdot}75 = 647{\cdot}50$ $647{\cdot}50 \times 0{\cdot}135 = 87{\cdot}41$ $647{\cdot}50 + 87{\cdot}41 = [€]\ 734{\cdot}91$	**Scale 10C (0, 4, 7, 10)** *Low Partial Credit:* • Work of merit, for example, shows some knowledge of percentages; some relevant multiplication *High Partial Credit:* • Substantial work of merit
(c)	$V = 100 \times 75 \times 65 = 487500$ [cm^3] $487500 \div 1000 = 487{\cdot}50$ [litres]	**Scale 10C (0, 4, 7, 10)** *Low Partial Credit:* • Work of merit, for example, identifies length, width, and height; multiplication of 2 relevant numbers; indicates division by 1000 *High Partial Credit* • $100 \times 75 \times 65$
(d) (i)	$08{:}45 + 2{:}38 = 11{:}23$	**Scale 5B (0, 2, 5)** *Partial Credit:* • Attempt at adding the relevant times

Q9	Model Solution – 45 Marks	Marking Notes
(d) (ii)	$54 \div 9 = 6$ $6 \times 8 = 48$ [students]	**Scale 5C (0, 2, 3, 5)** *Low Partial Credit:* • Work of merit • Identifies 9 parts *High Partial Credit:* • Calculates number in one part (6)
(e)	*Answer*: Graph B *Reason*: Slow and then fast	**Scale 5B (0, 2, 5)** *Partial Credit:* • Correct graph chosen • Incorrect graph with some work of merit in reason

Q10	Model Solution – 45 Marks	Marking Notes
(a)(i) **(ii)**	**(i)** $15{\cdot}80 \times 35 = [€]\ 553$ **(ii)** $15{\cdot}80 \times 0{\cdot}5 = 7{\cdot}90$ $15{\cdot}80 + 7{\cdot}90 = 23{\cdot}70$ **OR** $15{\cdot}80 \times 1{\cdot}5 = 23{\cdot}70$	**Scale 10D (0, 4, 6, 8, 10)** *Low Partial Credit:* • Work of merit, for example, in **(i)**, relevant multiplication; in **(ii)**, relevant multiplication; mentions $0{\cdot}5$ or $1{\cdot}5$ *Mid Partial Credit:* • One part correct • Work of merit in both parts *High Partial Correct:* • One part correct and work of merit in the other part
(a)(iii)	$671{\cdot}50 - 553 = 118{\cdot}50$ $118{\cdot}50 \div 23{\cdot}70 = 5$ [hours]	**Scale 5C (0, 2, 3, 5)** *Low Partial Credit:* • Work of merit *High Partial Credit:* • Finds overtime pay $(118{\cdot}50)$
(b) **(i)** **(ii)**	**(i)** $35000 \times 0{\cdot}2 = [€]\ 7000$ **(ii)** $7000 - 4000 = [€]\ 3000$	**Scale 10D (0, 4, 6, 8, 10)** *Low Partial Credit* • Work of merit, for example, in **(i)**, shows some knowledge of percentages; in **(ii)**, some relevant subtraction *Mid Partial Credit:* • One part correct • Work of merit in both parts *High Partial Correct:* • One part correct and work of merit in the other part

Q10	Model Solution – 45 Marks	Marking Notes
(b)(iii)	$35\,000 - (3000 + 1445 + 596)$ $35\,000 - 5041 = [€]\,29\,959$	**Scale 5C (0, 2, 3, 5)** *Low Partial Credit:* • Work of merit, for example, some relevant addition or subtraction *High Partial Credit:* • Finds total deductions • Correctly subtracts two deductions
(c)(i)	$50 \times 52 = [€]\,2600$	**Scale 5B (0, 2, 5)** *Partial Credit:* • Work of merit
(c)(ii)	Year 1: $2000 \times 0{\cdot}03 = 60$ $2000 + 60 = 2060$ $2060 + 1000 = 3060$ Year 2: $3060 \times 0{\cdot}03 = 91{\cdot}80$ $3060 + 91{\cdot}80 = [€]\,3151{\cdot}80$	**Scale 10D (0, 4, 6, 8, 10)** *Low Partial Credit:* • Work of merit, for example, shows some knowledge of percentages • Mentions $0{\cdot}03$ or $1{\cdot}03$ *Mid Partial Credit:* • Calculates value after 1 year (2060) *High Partial Credit:* • Calculates interest in year 2 $(91{\cdot}80)$ • Fails to include 1000 and correctly calculates compound interest for two years

Q11	Model Solution – 45 Marks	Marking Notes
(a)(i)	$6{\cdot}5 + 4 + 3{\cdot}5 + 5 = 19$ [m]	**Scale 5B (0, 2, 5)** *Low Partial Credit:* • Work of merit, for example, some relevant addition
(a)(ii)	$(0{\cdot}5 \times 3 \times 4) + (3{\cdot}5 \times 4)$ $= 6 + 14 = 20$ [m^2] **OR** $(6{\cdot}5 \times 4) - (0{\cdot}5 \times 3 \times 4)$ $= 26 - 6 = 20$ [m^2]	**Scale 10C (0, 4, 7, 10)** *Low Partial Credit:* • Work of merit, for example, divides area into rectangle and triangle; multiplication of two relevant numbers *High Partial Credit:* • Calculates area of triangle (6)
(b)(i)	Brand A: $20 \times 28 = £560$ Brand B: $26 \times 28 = €728$ $728 \times 0{\cdot}15 = 109{\cdot}20$ $728 - 109{\cdot}20 =$ [€] $618{\cdot}80$ **OR** $26 \times 28 = €728$ $728 \times 0{\cdot}85 =$ [€] $618{\cdot}80$	**Scale 10C (0, 4, 7, 10)** *Low Partial Credit:* • Work of merit, for example some relevant multiplication *High Partial Credit:* • Calculates cost of brand A (560) and brand B before discount (728) • Calculates cost of brand B including discount $(618{\cdot}80)$
(b)(ii)	*Answer*: Brand B *Calculations*: $560 \div 0{\cdot}87 =$ [€] $643{\cdot}68$ **OR** $618{\cdot}80 \times 0{\cdot}87 =$ [£] $538{\cdot}36$	**Scale 5C (0, 2, 3, 5)** *Low Partial Credit:* • Work of merit in calculations, for example, indicates division by $0{\cdot}87$ • Correct answer selected *High Partial Credit:* • Correct answer and work of merit in calculations • Incorrect or no answer selected and work of merit in calculations

Q11	Model Solution – 45 Marks	Marking Notes
(c)(i) **(ii)**	**(i)** 50 [cm] **(ii)** $\pi \times 50^2 = 7853{\cdot}98 = 7854$ [cm^2]	**Scale 10C (0, 4, 7, 10)** *Low Partial Credit:* • Work of merit, for example, in **(i)**, divides by 2; in **(ii)**, correct formula *High Partial Credit* • Part **(i)** correct and work of merit in part **(ii)**
(c)(iii)	$7854 \times 2 = 15\,708$ cent $15\,708$ cent $= 157{\cdot}08$ $157{\cdot}08 + 20 = [€]\ 177{\cdot}08$ **OR** $7854 \times 0{\cdot}02 = 157{\cdot}08$ $157{\cdot}08 + 20 = [€]\ 177{\cdot}08$	**Scale 5C (0, 2, 3, 5)** *Low Partial Credit:* • Work of merit, for example indicates multiplication of 7854 by 2 and stops *High Partial Credit* • Finds cost excluding additional charge $(157{\cdot}08)$